



Biochar as a Potential Nutrient Carrier for Agricultural Applications

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Abstract

Purpose of Review This paper offers a thorough overview of the processes of nutrient enrichment by biochar and how biochar as a nutrient carrier can effectively improve agricultural productivity. The distributions of nutrients in biochar and the dynamics of nutrients in soil are also reviewed in detail.

Recent Findings The application of biochar improves soil health by changing the soil's biological and physico-chemical characteristics of the soil, such as its structure, cation exchange capacity and microbial biomass carbon. Additionally, biochar produced from low pyrolysis temperatures can enhance nutrient retention in soils and be utilized as a carbon-based fertilizer.

Summary The maintenance of an adequate amount of organic matter in soil and a dynamic biogeochemical cycle of essential nutrients are key components of sustainable soil management. Biochar is a carbonized biomass derived from various feedstock materials, including wood and crop residues, manures, biosolids and animal carcasses. Biochar has been used for more than two decades as a soil amendment to improve soil physicochemical conditions and mitigate soil contamination. Nutrient-enriched biochar-based fertilizers (NEBBF) can be prepared using various nutrient enrichment procedures and have the potential to increase soil fertility and crop productivity. The application of NEBBF, which is a carbon-based nutrient source, has been shown to enhance microbial activity, thereby increasing the efficiency of nutrient use compared to conventional non-carbon-based synthetic fertilizers. This review identified key research gaps and discussed the importance and necessity of biochar as a nutrient carrier in agriculture.

Highlights

- Biochar is an ideal nutrient carrier due to its nutrient enrichment characteristics
- The co-occurrence of nutrients and carbon in biochar improves the nutrient use efficiency
- Biochar-based nutrient sources promote plant growth even in degraded soils
- Nutrient-enriched biochar aids in the cleanup of polluted soils
- Nutrient-enriched biochar can replace polymer-based slow-release fertilizers

Keywords Biochar · Nutrient-enriched biochar · Carbon-based fertilizer · Agricultural productivity · Nutrient carrier · Soil amendment

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Introduction

The farming sector and scientific community are paying more attention to the use of biochar as a soil amendment. Researchers have discussed biochar's economic viability and ecological advantages in the last decade, including their role in carbon sequestration and cleaning up polluted land [1–3]. The type of feedstock materials and the pyrolysis conditions usually affect the amount of nutrients in the biochar. For example, biochars made from feedstock materials based on manure and biosolids often have higher concentrations of nitrogen and phosphorus than biochars made from feedstock materials derived from straw and wood [4, 5]. An increase in ash content (inorganic residues) after pyrolysis causes an increase in the phosphorus (P), calcium (Ca), magnesium (Mg), silicon (Si) and potassium (K) contents of biochar, whereas nitrogen (N) content falls when pyrolysis temperature rises due to the gaseous emission [6–8]. Biochar often acts as a nutrient sink; therefore, it can hold nutrients, i.e. N, P and K, effectively; otherwise, losses of nutrients like N through leaching and gaseous emissions (nitrous oxides) have been reported by various researchers [9–11]. According to Yu et al. [12], the porosity and surface charge of biochar determines its capability to retain nutrients, i.e. the ability to retain cations and anions. Nonetheless, the pH of the biochar mostly regulates the nitrogen losses by ammonia emissions [13, 14].

Adding biochar affects several soil parameters, such as cation exchange capacity (CEC), pH, biological activity, water retention, bulk density and biological activity. The changes in soil properties might impact the microbial transformation of nutrients and nutrient interactions with soil particles [15–17]. By raising the soil's nutrient levels and improving the soil condition, biochar increases crop productivity and soil fertility. Therefore, it promotes aeration [18, 19] lowers bulk density [20], enhances microbial activity and water retention [21], buffers soil reactions [22] and preserves soil aggregate structure [23, 24]. According to Ducey et al. [25], biochar can alter the composition of the soil microbial populations and affect plant uptake and cycling of nutrients [26]. Nitrate leaching is decreased in soil due to biochar's ability to inhibit the nitrification process [27]. The agronomic and environmental applications of biochar for crop production, restoration of soil health, soil remediation and carbon sequestration have been discussed by several researchers [12, 22, 28–39]. Several reviews and meta-analyses with an emphasis on the dynamics of soil-nutrient exchange and biochar as a microbial carrier have also been published in the last decade [40–42]. There is no synthesis of data available on the usages of biochar as a nutrient carrier, which can be effective in improving agricultural productivity. The novel aspects of this paper can be described based

on the fact that this review focuses on (a) distribution and enrichment of nutrients in biochar; (b) dynamic of nutrients in soil derived from biochar and (c) value of agricultural application of nutrient-enriched biochar.

In order to improve soil quality and increase crop yield, nutrient-enriched biochar-based fertilizers (NEBBF) are very effective and must be applied to soils in balanced amounts [28–31]. Researchers have found that NEBBF is an effective soil amendment that allows a low-cost, intensive N application to the soil without any harmful effects on soil quality [41, 42]. Mineral-biochar composites made recently with waste, rock phosphate (RP), clay minerals, and mineral salts, mica, have emerged as a workable method to improve biochar's functional roles in agricultural applications [30]. Minerals are employed in a variety of soil and environmental applications and are integrated into the heterocyclic carbon chain of the biochar, which results in the enhancement of biochar properties through chemical fertilizers [25–27, 32, 33]. To achieve sustainable agricultural production, a deeper insight into the agricultural applications of NEBBF is necessary. Therefore, this review provides information on synthesis and properties, nutrient source and enrichment mechanisms and the importance of nutrient-enriched biochar-based fertilizers in agriculture. In addition, the functions of biochar and the production of biochar organo-mineral complexes are discussed in the review, along with their agro-environmental applications.

Methodology

The Web of Science, Scopus and Pubmed databases provided the literature data, which were then evaluated in accordance with the following search keywords: TS = ("biochar*") AND TS= ("nutrient enrich*" OR "fertilizer" OR "nutrient distribution" OR "nutrient dynamics" OR "mineral-enriched biochar" OR "biochar based fertilizer" OR "nutrient-enriched biochar" OR "biochar fertilizer" OR "pyrolysis process"). A total of 802 results were obtained, among which a total of 227 results were included for the present systematic review based on appropriate study methodologies. The software VOSviewer (1.6.19) was used to visualize the results and used for an in-depth study of selecting the papers that were subsequently reviewed (Fig. 1). Additionally, such network visualization bibliography analysis is used to graphically represent the relationships between key publications, research areas and authors. A PRISMA flow chart is used to screen the relevant literature as well as keywords/search terms concerning nutrient-enriched biochar-based fertilizers (NEBBF) (Fig. S1). It was found that within the past 10 years, there had been a sharp increase in the number of publications on the use of biochar as a nutrient carrier. Important topics covered in Fig. 1 include the

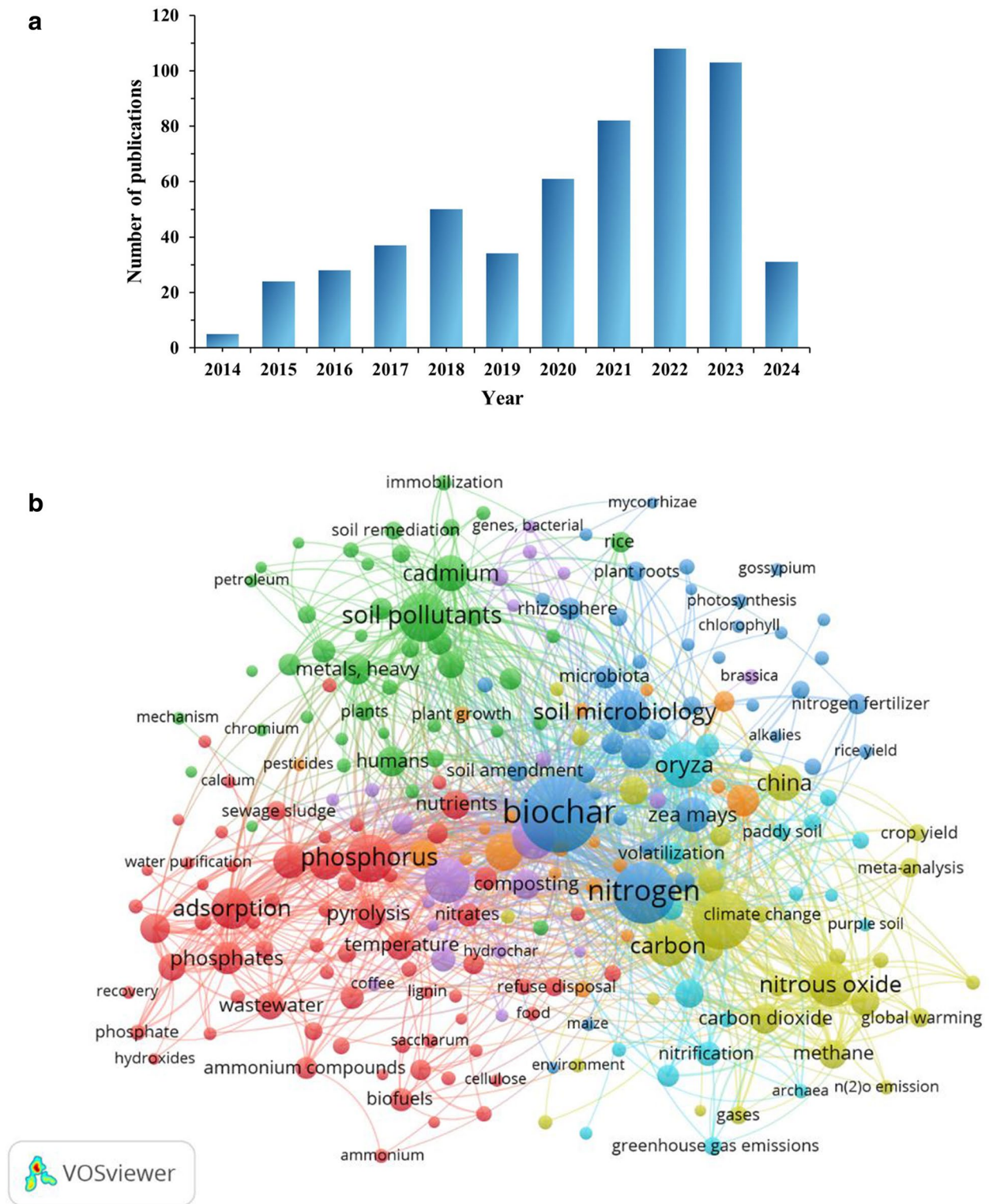


Fig. 1 **A** The number of publications over the last 10 years covering the aspects of the use of biochar as a nutrient carrier. **B** Keyword co-occurrence map showing the most frequently investigated topics using VOS viewer

distribution of nutrients in biochar, the dynamics of nutrients during the pyrolysis process, the enrichment of biochar with nutrients, the application of nutrient-enriched biochar for the delivery of nutrients and the enhancement of crop yield through NEBBF. The following sections contain a thorough explanation of the above-mentioned topics.

Synthesis and Properties of Biochar

Biochars can be produced using various methods; the production conditions can be tailored to optimize the yield, quality or properties of the resultant biochar (Table 1). These methods encompass a range of temperature regimes, heating rates, feedstock and reactor types. Understanding the diversity of synthesis techniques is crucial for tailoring biochar production to meet specific application requirements. Below, we provide an overview of the different synthesis methods employed in biochar production.

Torrefaction is one of the methods for producing biochars; this method involves the treatment of biomass in an oxygen-free atmosphere at comparatively low temperatures (200–300 °C) with a heating rate of $< 10\text{ }^{\circ}\text{C min}^{-1}$ and a residence period ranging from a few minutes to several hours [60, 61]. During torrefaction, volatile components such as water, acids and sugars are removed, leaving behind a dry, carbon (C)-rich material with increased energy density [62]. The resulting torrefied biomass, often called “biocoal”, can be used as a renewable energy feedstock in thermal power plants, industrial boilers and co-firing applications with coal [63, 64].

Slow pyrolysis involves gradual heating of biomass in between 350 and 800 °C in the absence of oxygen. This is typically performed at $< 10\text{ }^{\circ}\text{C min}^{-1}$, and the biomass can remain at these temperatures for hours to even days [65, 66]. This makes it possible to convert organic biomass materials into syngas, i.e. a mixture of hydrogen, carbon monoxide and methane, biochar and bio-oil under controlled conditions. Instead of producing gaseous and liquid products, slow pyrolysis usually promotes the production of biochar [60, 61]. One of the key advantages of the slow pyrolysis method is its ability to retain organic C and nutrients in the resulting biochar, thereby enhancing its value as a soil conditioner and nutrient carrier. This flexibility enables the production of biochars with tailored characteristics suited to specific soil types, crop requirements, environmental conditions or soil management strategies.

Fast pyrolysis represents a more rapid thermal decomposition process conducted at higher temperatures (300–1000 °C), with $10\text{--}200\text{ }^{\circ}\text{C s}^{-1}$ heating rates and shorter residence times ($< 30\text{ min}$) [66–68]. The biomass is rapidly heated to temperatures above its decomposition point, leading to the rapid release of volatile compounds, which are subsequently condensed into bio-oil, leaving behind biochar as a solid

residue. Fast pyrolysis offers several advantages, including higher bio-oil yields, shorter processing times, and reduced energy requirements compared to slow pyrolysis [69]. Fast pyrolysis is a promising technology for converting biomass residues and waste materials into biochar products for soil application, C sequestration and as renewable energy [70].

Gasification involves the partial oxidation of biomass at high temperatures ($> 500\text{ }^{\circ}\text{C}$), with heating rates ranging from $10\text{ to }30\text{ }^{\circ}\text{C min}^{-1}$ and a residence time of $\sim 10\text{--}20\text{ s}$ in a controlled environment, such as in a gasifier [65, 71]. This process produces syngas and biochar as byproducts. The syngas produced during gasification can be used for heat and power generation, hydrogen production or conversion into liquid fuels through Fischer-Tropsch synthesis [72]. Gasification technologies vary in complexity and scale, ranging from small-scale, decentralized systems for rural applications to large-scale, integrated facilities for industrial and municipal waste management [73]. Gasification-derived biochar exhibits unique properties, such as high surface area, porosity and stability, making it suitable for applications in soil remediation, C sequestration and wastewater treatment [56].

Wet biomass is treated using the hydrothermal carbonization (HTC) technique in an aqueous environment at temperatures between 350 and 600 °C, with a heating rate of $10\text{--}300\text{ }^{\circ}\text{C s}^{-1}$ in a residence period of more than 15 s, and a pressure of 5–20 MPa [68]. Under these conditions, biomass undergoes hydrolysis, dehydration and carbonization reactions (decarboxylation, polymerization and aromatization), forming a hydrochar, a solid carbonaceous material resembling biochar [74]. The HTC offers advantages in processing wet feedstocks with high moisture content, such as sewage sludge, food waste and agricultural residues, without needing pre-drying or high-energy inputs [75]. The HTC process can be conducted in batch or continuous systems, with varying reaction times and conditions to optimize hydrochar yield and quality. Hydrochar can be used as a soil amendment, filter media, or precursor for activated C production in environmental and agricultural applications [76]. Additionally, HTC offers opportunities to add values to organic waste streams, reduce greenhouse gas emissions and close nutrient cycles in circular bioeconomy models [77, 78].

Biochar exhibits diverse chemical and physical properties, influenced by its synthesis method, feedstock composition and processing conditions. Understanding and characterizing these chemical and physical properties are essential for optimizing biochar production processes and harnessing their full potential in sustainable agriculture and sustainable environmental applications, including remediation and regulation of climatic factors. The behavior and application of biochar in different sectors are mostly determined by its chemical properties, which include pH, C content, CEC and the availability of surface active functional groups. Multiple

Table 1 Summary of selected publications illustrating the impact of various pyrolysing conditions on nutrient contents of biochars derived from diverse feedstock sources

Production method	Feedstock	Production condition			Nutrient content							Cation exchange capacity cmol ₍₊₎ kg ⁻¹	References
		Pyro. temp. °C	Heat. rate °C min ⁻¹	Retention time	C	H	O	N	P	K	Ca		
Torrefaction	Macroalgae	275		30 min	545	53	196	58	12	27	11	7.4	Guo et al. [43]
	Grape marc				600	61	236	26	0.3	27	8.4	1.5	
	Wood				545	65	386	0.6	< 0.1	0.5	0.9	0.1	He et al. [44]
	Poplar wood	200		30 min	477	66	451	4					
		250			510	64	421	4					
Slow pyrolysis		300			622	58	314	5					Igalavithana et al. [45]
	Vegetable waste	200	7	2 h	529	69	360	42					
	Pinecone	200	7	2 h	697	21	271	10					Lu et al. [46]
	Vegetable + pinecone	200	7	2 h	547	59	389	5.7					
	Bamboo	750		3 h	860	15		4.5				15	
	Rice straw	500		30 min	508	17		17				45	Figueiredo et al. [47]
	Sewage sludge	300	11	30 min	230	36		33	4.1	1.1	6.7	1.8	
		400	11	30 min	215	27		29	5.4	1.1	7.1	1.7	Vimal et al. [48]
		500	11	30 min	190	16		23	6.1	1.3	7.8	1.7	
	Mixture of poultry litter, banana peduncle and phosphogypsum	700	10	1 h	193	5.5		3.8	24	51	53		

Table 1 (continued)

Production method	Feedstock	Production condition			Nutrient content							Cation exchange capacity cmol ₍₊₎ kg ⁻¹	References
		Pyro. temp. °C	Heat. rate °C min ⁻¹	Retention time	C	H	O	N	P	K	Ca	Mg	
	Oilseed rape straw	500	10	2 h	666	27	120	12	1.2		11		Chen et al. [49]
	Food waste digestate	400	10	2 h	469	21	147	19					Liu et al. [50]
		500	10	2 h	483	15	119	19					
		600	10	2 h	504	13	94	19					
		700	10	2 h	502	11	91	18					
		800	10	2 h	502	11	89	18					
	Switchgrass	500	15	2 h	394	13		7					Brewer et al. [51]
	Corn stover	500	15	30 min	628	29		13					
	Rice straw	250		2 h	572	-	-	11	1.3	169			Peng et al. [52]
				4 h	635			12	1.6	189			
				8 h	636			16	1.9	212			
		300		2 h	655			14	1.6	198			
				4 h	676			15	1.6	199			
				8 h	657			17	1.9	215			
		350		2 h	661			14	1.8	207			
				4 h	662			15	1.9	222			
				8 h	658			17	2.3	261			
		400		2 h	713			15	2.1	246			
				4 h	697			13	2.2	267			
				8 h	707			12	2.5	299			
		450		2 h	706			9.7	2.2	264			
				4 h	729			11	2.4	278			
				8 h	720			11	2.8	314			
	Mixture of paper sludge and wheat straw	500		20 min	511	17	109	14	0.8	10	62	3.2	Kalderis et al. [53]

Table 1 (continued)

Production method	Feedstock	Production condition			Nutrient content						Cation exchange capacity cmol ₍₊₎ kg ⁻¹	References	
		Pyro. temp. °C	Heat. rate °C min ⁻¹	Retention time	C	H	O	N	P	K			Ca
Fast pyrolysis	Switchgrass	500	15	1.5 s	387	25		6					Brewer et al. [51]
	Corn stover	500	15	1.5 s	378	25		8					
	Pine sawdust	400		3 s	517	50	338	9.3	12	69	230	-	Laghari et al. [22]
		500			547	39	213	8.6	14	83	376	-	25
		600			651	37	193	6.4	18	107	378	-	25
		700			736	30	129	6	21	119	468	-	24
Gasification		800			775	25	108	5.6	12	61	460	55	23
	Switchgrass	760	15		428	16		8					Brewer et al. [51]
	Corn stover	730	15		385	13		7					
	Coconut shells	850	20	1 h	792	13	113	1	0.3	20	0.4	0.3	Romero Millán et al. [54]
	Bamboo guadua				608	8	66	3	1.9	31	1.3	0.8	
	Oil palm shells				852	8	86	5	2.3	6.4	1.6	1.6	
Hydrothermal carbonization	Pine chip	900		10 s	793	12	85	0.2	1.3	9.3	21	2.1	Marks et al. [55]
	Pine wood	750			468	9.7	131		4	72	18	4.6	Hansen et al. [56] and Hansen et al. [57]
	Winter wheat straw	1200			653	6.3	89		3.4	25	52	5.9	Xu et al. [58]
	Food waste	230		70 min	631			65	14	4	29	1	
	Mixture of sugarcane bagasse and vinasse	180		60 min	571			58	13	18	48	3	
	Orange peel	230		13 h	562	42	116	34	47	28	22	27	Bento et al. [59]
		200		8 h	662	52	210	12	0.2	12	10	1.1	Kalderis et al. [53]

Pyro. temp. pyrolysis temperature, heat. rate heating rate

factors, including feedstock type, pyrolysis temperature and initial pH of the feedstock, determine biochar pH. For example, feedstocks with a higher ash content tend to produce biochars with higher pH values due to the presence of alkaline minerals [10]. The pyrolysis temperature can also affect biochar pH, with higher temperatures often resulting in biochars with higher pH values. For example, coconut fibre biochar produced through slow pyrolysis at 350, 400, 450, 500, 550 and 600 °C had pH values of 7.6, 8.3, 8.6, 9.8, 9.6 and 10.0, respectively [79]. The breakdown of organic compounds at elevated temperatures leads to the release of alkaline minerals (e.g. CaCO_3 and MgCO_3)/ash (containing inorganic alkalimetals), which results in an increase in the pH value [80]. Additionally, high pyrolysis temperatures can promote the conversion of acidic functional groups ($-\text{COOH}$ and $-\text{OH}$) present in the biomass feedstock into more stable and less acidic forms, further contributing to higher pH values in the resulting biochar [81, 82]. However, CEC is influenced by various other factors, including biochar's surface area, pore structure and elemental composition. Biochars with higher surface areas typically exhibit higher CEC values, providing more cation exchange sites [10, 83].

Increasing the pyrolysis temperature of maize straw biochar from 300 to 400, 500, 600, 700 and finally to 800 °C resulted in a progressive increase in surface area, with values recorded at 1.33, 2.12, 26.51, 255.34, 451.87 and 668.63 $\text{m}^2 \text{g}^{-1}$, respectively [67, 84]. Similarly, micropore volume also displayed a corresponding rise across the temperature range, measured at 0.001, 0.001, 0.005, 0.095, 0.168 and 0.226 $\text{cm}^3 \text{g}^{-1}$, respectively, under those pyrolysis temperatures [84]. The positive relationship observed between micropore volume and surface area indicates that the distribution of pore sizes plays a vital role in increasing the surface area of biochar. Additionally, biochar particle size plays an important role in controlling soil properties. When biochar is added to light-textured sandy soils, soil aggregates are changed and the larger aggregates become more stable, increasing the soil's total pore volume and decreasing its bulk density. The larger generalized soil structure index (GSSI) showed that soil structure was greatly improved as a result of the application of the various biochar particles, which decreased bulk density and enhanced porosity [84, 85]. Furthermore, the higher surface area of the smaller-sized biochar enhances the forces of adhesion and cohesion between water molecules and biochar particles, increasing the soil's capacity to hold onto more water. The hydraulic and physical characteristics of soil are also impacted by the size of the biochar particles. Researchers have found that addition of larger biochar particles to sandy soils reduced their available water content (AWC) and hydrophobicity while increasing their total porosity (TP) and permanent wilting point (PWP) [77, 81, 85]. Verheijen et al. [85] have reported through a long-term incubation experiment that smaller biochar particles (0.05–1

mm) significantly decreased the bulk density of sandy soil, whereas larger particles (2.00–4 mm) had a stronger effect on reducing the bulk density of a sandy loam soil. Moreover, the application of biochar with small particle sizes at 20% (v/v) resulted in a 60% increase in gravimetric water holding capacity.

Biochar as a Nutrient Source

The primary advantage of biochar lies in its dual function as a nutrient source for both soil and plants, serving as a reservoir for essential elements such as N, P and K and releasing them gradually to plants based on their requirements, thereby minimizing nutrient losses and ensuring optimal supply to plant roots at the right time [86, 87]. The elemental composition of biochar (significantly variable due to factors such as feedstock type and pyrolysis conditions) influences its nutrient content and availability [80, 83]. Chan and Xu [88] and Lehmann and Joseph [89] reported variations in carbon (C 172–905 g kg^{-1}), nitrogen (N 1.8–56.4 g kg^{-1}), total phosphorus (P 2.7–480 g kg^{-1}) and total potassium (K 1.0 to 58 g kg^{-1}). Table 2 provides a comprehensive overview of nutrient dynamics in biochar production, summarizing key findings from various studies exploring the impact of feedstock type and pyrolysis temperature on nutrient content. This information is beneficial in providing guidelines on generating biochar containing more nutrients capable of being released into the soil as well as to have other noteworthy attributes of biochar that help to enhance soil quality (as a way of increasing C storage, improving soil physical and chemical properties). For example, adding other organic soil amendments, such as manure or compost, would also release nutrients into the soil but converting those into biochar have multiple environmental benefits. These include destruction of pathogens, organic contaminants and other potentially toxic elements present in the manures, improvement in plant growth and soil physico-chemical properties, enhancement of soil organic carbon and nutrient availability to the plants. Increased estrogen exposure in soils, i.e. stemming from direct application of manure may have a negative effect on soil microbial activity. The total nutritional balance from animal excreta to plant growth should be comparable to traditional manure management, even though it is anticipated that the environmental impact caused by manure nutrient losses will be reduced through biochar conversion. Furthermore, a portion of the carbon from labile manure would be transformed into a long-term carbon sink (non-easily mineralizable carbon).

In terms of biochar formation, the procedure involves significant changes in chemical composition. With higher torrefaction (pyrolysis) temperatures and prolonged residence time, carbon content increases in addition to the reduced hydrogen concentrations as a result of vaporization

Table 2 Summary of nutrient dynamics in biochar production

Nutrient released	Methodology and materials	Key findings	References
Macro elements (C, N and P), Ca, Mg, K and Na	Pyrolysis of seaweed species <i>Laminaria pallida</i> and <i>Gracilariopsis funicularis</i> at different temperatures	<i>G. funicularis</i> biochar had the highest macro element content (total C 38.3%, N 4.3% and P 6.3 g/kg) and cations (Ca, Mg, K and Na). Pyrolysis temperature of 400 °C produced optimal biochar quality.	Katakula et al. [90]
N, P and K exchangeable cations (K)	Pyrolysis of various brown and red seaweeds	Seaweed biochar has low C content and surface area, high yield and abundant essential trace elements. pH ranges from neutral to alkaline, suitable for diverse soils.	Roberts et al. [87]
Varies depending on feedstock and temperature	Pyrolysis of various organic waste materials at different temperatures	Pyrolysis temperature and feedstock type significantly influenced biochar properties like pH, surface area, cation exchange capacity, ash content, fixed carbon, degree of aromaticity, and mineral composition. Higher temperatures (550 and 650 °C) produced biochar suitable for carbon sequestration and agriculture.	Pariyar et al. [83]
N (decreased), P and K (increased)	Pyrolysis of wheat straw and rice straw at 300, 400 and 500 °C	Biochar yield decreased with increasing temperature. Ash content and pH increased with increasing temperature. N content decreased, while P and K contents increased with increasing temperature. Biochar from rice straw had higher P and K contents at 500 °C. Selected feedstock and low pyrolysis temperature may produce nutrient-rich biochar.	Naeem et al. [86]
P (reduced)	Compost with varying C/P ratios from different feedstocks	Higher compost pH (due to feedstock composition) reduced readily available P. Adding biochar further reduced readily available P, especially with biochar high in Ca, Fe and Al.	Vandecasteele et al. [91]
N, P, K and Mg (varied with feedstock and temperature)	Jatropha non-oil biomass (leaf, trunk and branch) pyrolyzed at 300, 400, 500 and 600 °C	Leaf biochar had lower carbon and higher N, P, K and Mg than other biochars. Trunk biochar had the highest calcium content. All biochars, except leaf biochar at 300 °C, improved water holding capacity. Diverse biochars with various properties can be tailor-made for specific needs.	Konaka et al. [92]
P, Ca, K, Na, Fe, Zn, Cu and Mn (increased with temperature)	Biochar from various feedstocks pyrolyzed at 300, 400, 500 and 600 °C	Biochar yield decreased with increasing temperature. pH and available element concentrations (except P in walnut shell) increased with increasing temperature. Biochars at 300 and 400 °C may be suitable for calcareous soils due to low pH and EC.	Reyhanitabar et al. [93]
N, P, K, Ca, Mg and S (varied with temperature and feedstock)	Biochar from corn cob, poultry litter, cow dung, and peanut shell pyrolyzed at 300, 400, 500 and 600 °C	Highest nutrient concentrations observed at 300 °C for most elements. Animal-derived feedstocks had higher nutrient content than plant-derived feedstocks. Increasing temperature decreased most nutrient concentrations and increased pH and ash content.	Omotade et al. [94]

Table 2 (continued)

Nutrient released	Methodology and materials	Key findings	References
P, K, Ca, Mg, Fe, Mn and Cu (increased with temperature)	Pecan branches, leaves and nut shells pyrolyzed at 300, 400, 500 and 600 °C	Biochar yield decreased with increasing temperature. Leaf biochars had higher surface area and lower average pore diameter. pH, CEC and water holding capacity (except for nut shells at 500 and 600 °C) increased with temperature. Higher nutrient content observed in branch and leaf biochars, especially K and N. Higher AP, AK, Ca, Mg, Fe, Mn and Cu at higher temperatures.	Liu et al. [95]
P, K (increased) and N (decreased)	Maize straw biochar at 300, 400 and 500 °C	CEC, pH, EC and carbon content increased with temperature. P, K, Ca and Mg increased with temperature; N decreased. AB-DTPA extractable nutrients decreased with temperature. Biochar at 300 and 400 °C increased maize growth, while 500 °C decreased it. P and K concentrations in maize increased with biochar and fertilizer. Biochar at low temperature may be best for crop growth.	Naeem et al. [96]
K (water-soluble)	Rice straw, wheat straw, maize stover and pearl millet stover biochar at 400, 500 and 600 °C	PSB biochar had the highest pH, CCE and carbonate content. Pyrolysis temperature increased pH, CCE and CO ₃ content. All biochars had high water-soluble K. Different residue types and temperatures produced biochars with diverse properties for specific soil needs.	Bera et al. [97]
N (decreased), P (no change) and K (no change)	Giant reed biochar at 300–600 °C	N loss and transformation increased with temperature. P was less soluble at higher temperatures. P and K release favoured by pH ≤ 5. Low temperature biochars (≤ 400 °C) had good N and P adsorption. Low temperature may be optimal for biochar from giant reed to improve nutrient availability.	Zheng et al. [98]
All elements	Various feedstocks and pyrolysis temperatures	Feedstock type influenced biochar properties more than temperature. Higher temperature increased fixed carbon and cation exchange capacity but decreased dissolved organic carbon and polarity index. All biochars had high boron content. Selecting feedstock and temperature can tailor biochar for specific environmental applications.	Das et al. [99]

Table 2 (continued)

Nutrient released	Methodology and materials	Key findings	References
P (increased)	Canola stalk and rice straw biochar at 350, 450, 550 and 650 °C	Total P and Olsen-P increased with temperature and were higher in canola stalk biochar. Biochar increased soil P content (depending on temperature, feedstock and soil type). Highest P availability at 450 °C biochar addition. Biochar increased Al-P and Fe-P in soils. Biochar pH positively correlated with Olsen-P and inorganic P. Choosing optimal pyrolysis temperature is crucial for improving soil P availability.	Yang and Lu [100]
P (varied with extractant and temperature)	Sewage sludge biochar at 200, 300, 500 and 600 °C	Pyrolysis temperature increased pH, surface area, pore volume, ash, fixed carbon, Ca, Mg, Zn and total P. Solubility of biochar-P varied with extractant and temperature. P-NAC solubility reached a maximum at 300 °C and then decreased. P-CA and available P only affected at 600 °C. Considering different P solubilities in extractants is recommended when using biochar as P fertilizer.	de Figueiredo et al. [101]
N, P and K (decreased at high temperature)	Domestic sewage sludge biochar at 260, 420 and 610 °C	Higher temperature decreased VOM, TN and P but increased pH, EC, Na and K. The trace metal levels varied with temperature, with the lowest at 420 °C. Biochar at 420 °C had the lowest root elongation inhibition, suggesting it as the ideal temperature for safe soil amendment.	Zoghلامي et al. [102]
N and P (increased with poultry manure and slower heating)	<i>Gmelina arborea</i> sawdust and <i>Tithonia diversifolia</i> shoot biochar with or without poultry manure at 350 °C with heating rates of 5, 10 and 20 min	Poultry manure and slower heating rates significantly increased biochar yield and nutrient quality. Higher heating rates increased ash content. N and P contents varied with feedstock and heating rate. Reduced organic C, C/N and C/P influenced biochar yield and ash content. Poultry manure with sawdust or shoot under slow pyrolysis produced higher biochar yield and quality.	Oyeyiola et al. [103]

AP available phosphorus, AK available potassium, VOM volatile organic matter, TN total nitrogen

compounds [104]. The pyrolysis temperature plays a crucial role in shaping the biochar composition, as shown by several studies on *Chlorella vulgaris* and torrefied biomass, where the amounts of C, H and N were found higher in charred biomass due to the protein depolymerization process, which leads to the release of mostly non-carbon elements [105–107]. The torrefaction process led to the development of a more hydrophobic aromatic hydrocarbon structure with a low O/C ratio. It decreased the oxygen-containing compounds resulting from the extensive thermochemical conversion of cellulose, hemicellulose and lignin. These studies emphasized how pyrolysis conditions influence biochar characteristics, showing that temperature parameters play an important role in the tailored production of biochar to serve dual functions, such as providing both plant and soil nutrients (Fig. 2).

The types of biomass used in biochar production significantly influences its composition, with brown seaweed having higher carbon and hydrogen contents than red seaweed and varying in sulphur, potassium, phosphorus and nitrogen content [87]. This highlights the importance of considering the elemental composition of the starting biomass, as it directly influences the properties of the resulting biochar. Arguably, the C/N ratio becomes critical in determining

biochar's viability as soil fertility. This ratio is of great importance in the evaluation of organic matter decomposition as well as nutrient dynamics within soils. Research conducted on *Chlorella vulgaris* FSP-E dried biomass and in other studies have shown that higher C/N ratios are in some biochar types, such as those derived from algal sources [107]. Such an increased C/N ratio implies relatively high carbon content on a nitrogen basis and may affect soil fertility. Specifically, algal biochar has high contents of exchangeable inorganic elements like aluminium, calcium, iron, magnesium and potassium. These elements contribute to the biochar's capacity to enhance soil fertility, as demonstrated in studies, including those focused on *Chlorella vulgaris* [105]. Algal biochar, with its complex relationship between C/N ratio and essential inorganic elements abundance, has potential as a valuable soil amendment, enhancing farmland fertility for sustainable agricultural programs.

However, challenges persist in comprehending the actual impact of pyrolysis temperature on nutrient availability in biochar. Notably, Katakula et al. [90] have reported an overall increase in total P and N contents in biochar with higher pyrolysis temperatures. They also observed a decrease in extractable P and N, indicating a complex relationship between pyrolysis conditions and the

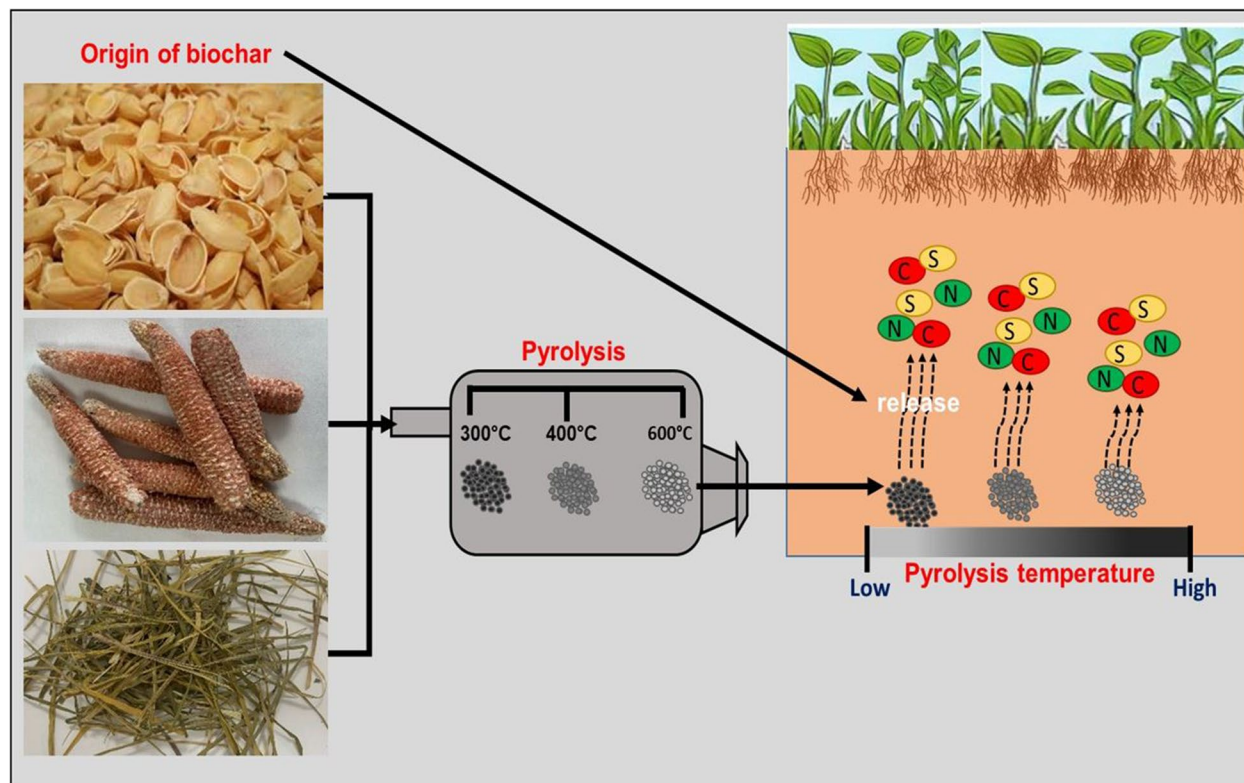


Fig. 2 Schematic diagram of the origin of biochar and the effect of pyrolysis temperature on the release of nutrients from biochar for plants

actual availability of these nutrients in a form accessible to plants. To address this discrepancy, Godlewska et al. [108] proposed alternative methods, such as vermicomposting, to enhance the availability of these nutrients in biochar. Vermicomposting, a process involving using earthworms to decompose organic matter, has been shown to be promising in improving nutrient accessibility in various organic materials. Understanding how vermicomposting or similar approaches can potentially modify the nutrient dynamics of biochar becomes imperative in optimizing its application as a soil amendment which stands out with higher P and K contents compared to other feedstocks at lower temperatures. However, this advantage comes with increased biochar yield, ash content and pH, coupled with a decrease in N content as the pyrolysis temperature rises [86]. One possible mechanism that can explain the higher P and K contents of rice straw biochar at 500 °C is the release of volatile matter from the biomass during pyrolysis. Volatile matter is the fraction of biomass that can be vaporized at high temperatures, and it mainly consists of carbon, hydrogen, oxygen and nitrogen [109]. As the pyrolysis temperature increases, more volatile matter is released, which reduces the mass of the biochar and increases its ash content [110]. Therefore, the higher ash content of rice straw biochar at 500 °C implies a higher concentration of P and K in the biochar.

The study of *Jatropha* non-oil biomass reveals variations in biochar nutrient content between plant tissues (trunk, branch and leaf), i.e. the trunk was enriched with Ca, whereas in leaf N, P, K and Mg concentrations were found to be higher than other parts. This suggests the potential to customize biochar nutrient content for specific agronomic needs by selecting different tissue types and adjusting pyrolysis temperatures, as with increasing temperature, a gradual increase in pH and electrical conductivity was observed [92, 93]. Moreover, Omotade et al. [94] investigated biochar from diverse feedstocks (poultry litter, peanut shell, cow dung, corn cob) across various pyrolysis temperatures (300–600 °C). They revealed that nutrient concentrations, i.e. Ca, N, K, P, S and Mg decreased as pyrolysis temperature increased with peak nutrient contents observed at 300 °C. They noted that animal-derived feedstocks had higher nutrient content than plant-derived ones. They highlighted the complex interplay of feedstock type and pyrolysis temperature on nutrient variability, where increasing temperature decreased most nutrient concentrations while elevating pH and ash content.

Furthermore, Liu et al. [95] explored pecan biochar properties, focusing on nutrient content variations across different pecan feedstocks and pyrolysis temperatures (300–600 °C). Similarly, Naeem et al. [96] investigated maize straw biochar at different temperatures, revealing that increasing pyrolysis temperature enhanced EC, pH,

nutrient concentrations (P, K, Ca and Mg) and CEC. Shoot and root dry matter of maize increased significantly with biochar from lower pyrolysis temperatures, emphasizing its potential for crop growth improvement. Another aspect of biochar variability is highlighted by Bera et al. [97], who assessed biochar from major Indian agricultural residues at varying pyrolysis temperatures. The study emphasized the profound influence of biochar temperature on soil properties such as pH, calcium carbonate equivalent (CCE) and water-soluble potassium content. The results suggested tailored biochars with diverse nutrient profiles based on residue type and pyrolysis temperature, addressing specific soil nutrient deficiencies. Moreover, Zheng et al. [98] explored giant reed biochar properties at different pyrolysis temperatures. The study indicated that low temperatures (≤ 400 °C) were optimal for nutrient availability, showcasing appreciable adsorption capacity for N and P. This finding suggests that biochar from lower pyrolysis temperatures can enhance nutrient retention in soils.

Das et al. [99] quantified the influence of pyrolysis temperature and feedstock types on biochar properties. The research highlighted the feedstock's dominant role, with increasing pyrolysis temperature impacting soil structural modification. The study emphasized the potential of biochar, produced under specific conditions, to address soil nutrient deficiencies and contribute to environmental applications. Yang and Lu [100] observed that biochar from canola stalk and rice straw exhibited variations in total phosphorus (P), available P and inorganic P with pyrolysis temperature, which might be attributed to the P preservation (high residual phosphate) event due to the carbon volatilization resulting from the decomposition of biomass at high temperature. The addition of biochar influenced P availability in soils, with the highest availability observed at 450 °C. Soil pH, biochar pH and inorganic P showed positive correlations in acidic soils. The effect was significant in acidic soils, though a non-significant effect was observed in alkaline soils. In alkaline soils, P solubility is mostly controlled by its interaction with Ca^{2+} , whereas in acidic soils, Al^{3+} and $\text{Fe}^{2+}/\text{Fe}^{3+}$ primarily control plant availability of P through forming Al- and Fe-phosphates. The P sorption onto Fe and Al oxides decreases with the addition of biochar to acidic soils, which, on the other hand, raises the pH of the soil. Additionally, biochar itself may contain calcium and magnesium depending on the initial feedstocks and pyrolysis process, which may impact the availability of P. Pyrolysis temperature and feedstock type played crucial roles in determining the biochar's P content and its impact on soil availability. Similarly, de Figueiredo et al. [101] investigated P solubility in sewage sludge-derived biochar (SSB) at different pyrolysis temperatures. Extractable P varied with temperature, affecting available phosphorus-neutral ammonium citrate (P-NAC) content and available P. The study emphasized the importance of

considering P solubility in various extractants when utilizing SSB as an alternative to inorganic P fertilizers.

Researchers have found that pyrolysis temperature significantly influences biochar characteristics [101, 102]. They suggested that 420 °C was the ideal temperature for producing biochar with higher fertility levels and quality standards for soil amendment. Similarly, Oyeyiola et al. [103] also found that blending poultry manure with *Gmelina arborea* sawdust during slow pyrolysis (350 °C) significantly improved biochar nutrient quality and yield. The study highlighted the potential for poultry manure to enrich the biochar with nutrient contents (N and P) and enhance biochar production with specific biomass combinations and slow pyrolysis conditions [103]. Seaweed/algal biochar is highlighted for its higher levels of essential plant macronutrients (Ca, P, K and N) compared to land-based biomass [90, 111]. While biochar amendments can increase vital plant nutrients like N and P, monitoring sodium (Na) concentrations in biochar can also be crucial [87]. Strategies such as co-composting with organic materials may mitigate the impact of high Na concentrations in biochar over time, which might be attributed to high CEC of the biochar stemming from the adsorption of organic matter, which might increase the CEC by displacing Na^+ with Ca^{2+} ions from soil solution [112]. While co-compost treatments altered soil CEC in accordance with the substitution of high alkaline biochar for low alkaline biochar, biochar treatments demonstrated enhanced soil CEC based on biochar's own CEC value [87, 112]. Soil CEC was effectively raised by reducing the crop residues to biochar ratio in co-composts. By altering the pH and raising the concentration of cations (such as Ca^{2+} , Mg^{2+} and K^+) on soil exchangeable surfaces, liming material or biochar could improve the CEC of acid soils as discussed earlier. As previously mentioned, biochar co-composts increase soil CEC through cation displacement mechanisms; the effect is less pronounced in biochar co-composts compared to the pristine biochar because of the dilution effect stemming from feedstock materials of the biochar co-compost treatments [101, 112].

Nutrient Enrichment of Biochar

Material Used to Produce Nutrient-Enriched Biochar-Based Fertilizers

There are several techniques to enrich biochar with nutrients to produce biochar-based fertilizers. In some cases, a combination of pre and post-pyrolysis is used to produce NEBBF. The feedstocks used to enrich biochar can be grouped into (i) chemical fertilizers, (ii) organic wastes, (iii) rock dust, (iv) minerals and (v) pure inorganic compounds. In the enrichment process, biochars are produced from a wide variety of biomass, including residues of crop, tree, mushroom

substrate and biogas substrates; industrial and domestic sludges; and several types of manure (Table 3).

Due to the ease of providing a specific nutrient or group of nutrients, conventional chemical fertilizers are the most used materials in biochar enrichment. The main chemical fertilizers used for the production of NEBBF are mono-ammonium phosphate, diammonium phosphate, urea and potassium chloride (Table 3). Typically, enrichment techniques aim to add nutrients to make biochar a fertilizer that is balanced with plant nutrients. Furthermore, in many cases, enrichment serves to produce slow nutrient-release NEBBF. For example, in the case of sewage sludge biochar, enrichment with K has a dual purpose, i.e. (a) to increase the concentration of this nutrient and (b) to slow its release into the soil, avoiding losses through leaching [127, 128]. When focusing on NEBBF with natural sources, non-industrialized materials are used in the enrichment, such as rock dust and specific minerals (apatite, bentonite and others) and residues from nutrient-rich plants (e.g. banana peduncle).

Most NEBBF are produced from mixing feedstocks. In this case, organic or mineral sources are mixed with the biomass in pre-pyrolysis followed by enrichment of the biochar with organic or mineral materials. In general, this procedure aims to supply a nutrient or group of nutrients and create properties in the NEBBF aiming to (a) reduce the nutrient release time [127, 128]; (b) reduce gas volatilization/emission [116, 129]; (c) increase nutrient solubilization/availability [130]; (d) increase nutrient use efficiency [131, 132]; and (e) reduce the availability of contaminants [133].

Techniques and Technological Routes Used to Produce NEBBF

NEBBFs can be produced using different procedures or technological routes. In general, the techniques used to produce NEBBFs can be grouped into pre-pyrolysis, post-pyrolysis or a combination of these two processes (Fig. 3). When enrichment occurs in pre-pyrolysis, feedstocks can be mixed by direct treatment or pre-treatment [134].

Bone, microalgae residue, mushroom residue compost, bacterial biomass waste, banana peduncle biomass and poultry litter are examples of feedstocks used to produce NEBBF throughout the direct treatment/pre-pyrolysis process. Afterwards, the mixture formed in pre-pyrolysis is subjected to pyrolysis. After pyrolysis, NEBBF is obtained directly or after the granulation/pelletization process. In some cases, biochar is also subjected to a post-pyrolysis procedure [134, 135].

In the post-pyrolysis technique, enrichment only occurs after biochar production. In this case, biochar is mixed with nutrient-rich materials through direct physical mixing, impregnation in different types of solutions or through an encapsulation process. Finally,

Table 3 Selected articles covering the diversity of methods for NEBBF production and their main characteristics and effects on the soil/plant

Biochar biomass	Enrichment type	Group of feedstocks	Types of fertilizer	Nutrients	Granulometry	Characteristics of NEBBF	Effect on soil/plant	References
Sewage sludge	Post-pyrolysis	Chemical fertilizer ¹	Potassium chloride	K	Granule/pellet	More efficient than conventional fertilizer	NEBBF promoted greater tuber production and greater accumulation of nutrients in the soil.	Fachini et al. [113]
Wheat straw	Post-pyrolysis	Chemical fertilizer	Diammonium phosphate	P and N	-	Slow-release fertilizers/residual fertilization	NEBBF reduced the P inputs in the form of inorganic fertilizer/residual effect.	Wali et al. [114]
Olive tree pruning	Post-pyrolysis	Chemical fertilizer ²	Urea/ammonium sulphate	N	Powder	Slow-release fertilizers		Castejón-del Pino et al. [115]
Paper fibre sludge/ grain husks	Post-pyrolysis	Chemical fertilizer	Ammonium sulphate	N	Powder	Potential to be used as fertilizers and decrease CO ₂ emissions.	NEBBF decreased CO ₂ emissions compared to compost, fertilizer, or their mixture without biochar/would provide the necessary available N for plant uptake.	Rizhiya et al. [116]
Landscaping wood	Post-pyrolysis	Commercial liquid fertilizer/organic waste + saccharose solution	NPK solution/fresh foliar tree and grass biomass + sugar beet	N, P, and K		Potential to be used as fertilizers.	NEBBF can improve the performance of <i>T. cocoa</i> plant growth and vigour beyond the levels achieved with either pure mineral fertilization or pure biochar.	Meyer zu Drewes et al. [117]
Corn cob	Pre-pyrolysis + post-pyrolysis	Chemical fertilizer + aqueous hydrogels	Triple superphosphate + carboxymethylcellulose	P	-	Super adsorption capacity/slow-release fertilizers	Cumulative P release ratio of 61% after 28 days of application	Zang et al. [118]
Wheat straw	Pre-pyrolysis	Chemical fertilizer + mineral + rock dust + salt	Urea + bentonite clay + rock phosphate	N, P, and S	-	More efficient than conventional fertilizer	NEBBF increased plant biomass by 67%, N uptake by 40%, and P by 46% compared to mineral fertilizer.	Chew et al. [119]
Rice straw/maize stove/Gram residue	Post-pyrolysis	Chemical fertilizer + organic waste ³	Urea + crop residues + fresh cow manure	N		Promising liming material	NEBBF increased the soil pH and CEC by 36% and 32%, respectively, showing comparable effects to agricultural lime.	Nain et al. [120]
Rice straw	Post-pyrolysis	Chemical fertilizer + mineral + organic waste	Potassium sulphate + dolomite + shell powder	K and S	Powder		The application of NEBBF is conducive to reducing N ₂ O emissions.	Zhao et al. [121]

Table 3 (continued)

Biochar biomass	Enrichment type	Group of feedstocks	Types of fertilizer	Nutrients	Granulometry	Characteristics of NEBBF	Effect on soil/plant	References
Residues of grape	Post-pyrolysis	Organic waste + rock dust	Cow manure + rock phosphate	P			NEBBF higher nutrient concentrations in the soil and improved nutrient availability in saline conditions.	Moradi et al. [122]
Poultry litter	Pre-pyrolysis	Organic waste + industrial waste	Banana peduncle + phosphogypsum	K, S, and P	Powder	Slow-release fertilizers		Vimal et al. [48]
Apple and grape tree pruning remains	Pre-pyrolysis	Rock dust ^{2,3}	Rock phosphate	P		Potential to be used as fertilizers.	NEBBF obtained 8.5 to 46% more yield from dry matter compared to chemical fertilizer.	Mousavi et al. [123]
Wheat straw	Post-pyrolysis	Mineral ²	Goethite		Powder	Highly effective and sustainable multi-functional material	Showed the synergistic effects of heavy metal pollution control and soil carbon sequestration.	Wang et al. [124]
Cotton straw	Pre-pyrolysis	Salt + mineral ⁴	Magnesium phosphate+ bentonite	P and Mg	Granule	Slow-release fertilizers and high degradability		An et al. [125]
Chipped hemp fibre	Pre-pyrolysis	Minerals	Hematite + melanterite + dolomite	Fe, S, Ca, and Mg		Enriched biochar substrate	NEBBF is a suitable on substrate for P removal and retention.	Boltet al. [126]

NEBBF nutrient-enriched biochar-based fertilizers

¹Use of corn starch as a binder

²Use of activators (nitric acid, phosphoric acid, potassium hydroxide, and hydrogen peroxide)

³Use of microorganisms

⁴Use of coating materials

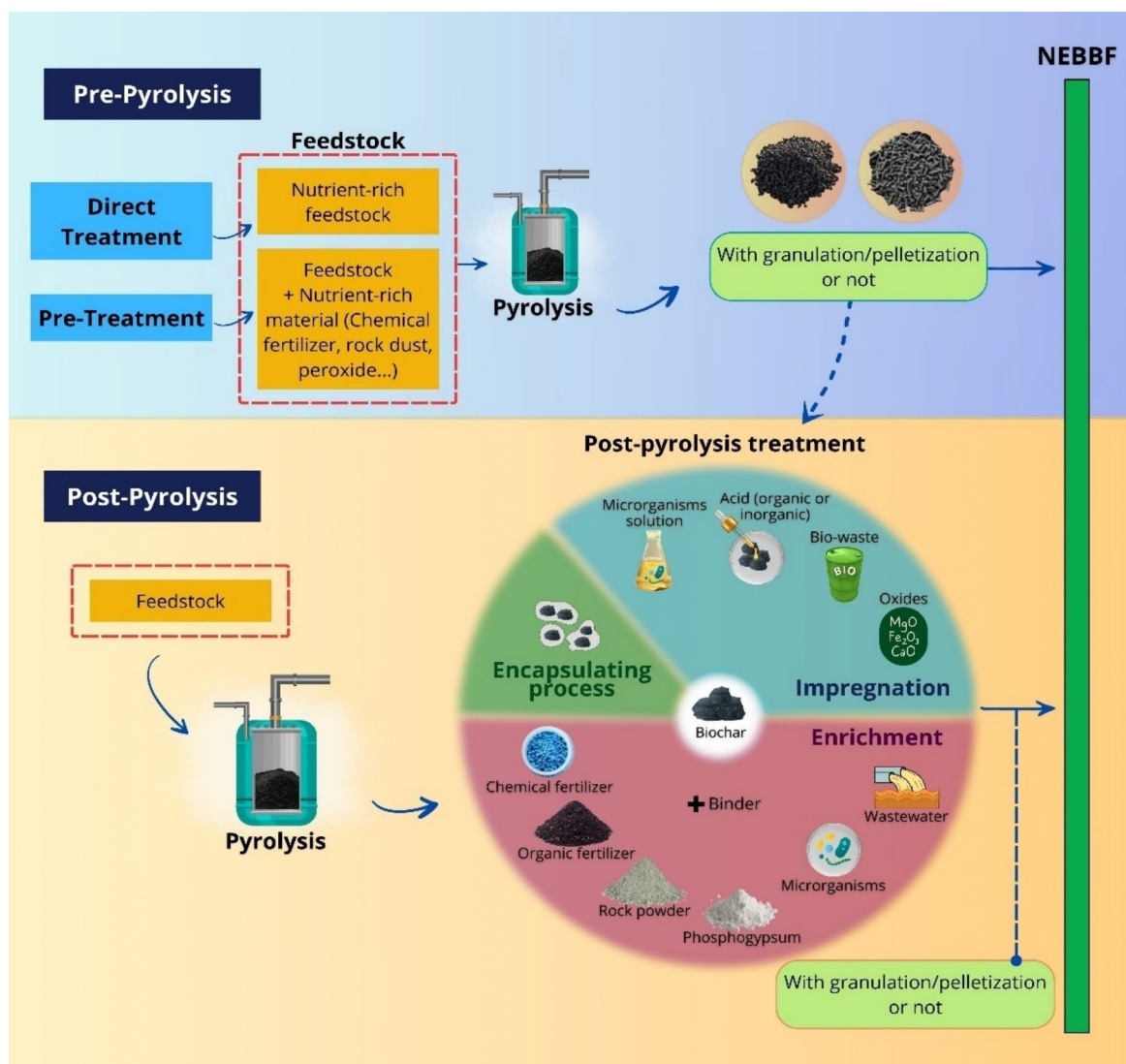


Fig. 3 Schematic diagram illustrating biochar enrichment techniques and NEBBF production

the enriched biochar may or may not be subjected to the granulation/pelletization process to obtain NEBBF. In direct post-pyrolysis enrichment, biochar is mixed with materials such as manure, clays, compost, chemical fertilizers and different nutrients under controlled or room temperature conditions to prepare NEBBF. Chemical fertilizers are the most used materials in biochar enrichment as sources of plant nutrients. Other bio-waste and microorganism solutions can also increase impregnation efficiency. The encapsulating process can be used to improve the delivery form of the enriched nutrient. Encapsulation can be carried out in biochar or even in NEBBF [135].

Agricultural Application of Nutrient-Enriched Biochar

Nutrient Retardation in Agricultural Applications of NEBBF

NEBBF has been adopted in agriculture in recent years due to its remarkable efficacy [136–138]. NEBBF addresses the nutrient deficiencies inherent in pure biochar and exhibits a controlled release of nutrients, contrasting with the rapid release characteristic of mineral fertilizers [134, 139, 140]. Furthermore, NEBBF holds substantial promise for augmenting soil health, carbon sequestration and enhancing both crop yields and nutritional quality [141–144].

Incubation experiments have unequivocally elucidated the nutrient release kinetics of NEBBF [128, 132], with a study indicating a 77% reduction in K release capacity compared to conventional mineral fertilizers [127]. The mechanistic understanding behind this slow potassium release dynamics might be attributed to the entrapment of nutrients by various pore volumes of biochar and different particle sizes. They have also explained that more soluble fractions/cations of NEBBF were greatly adsorbed through the carboxylic and phenolic surface functional groups of biochar. Roy et al. [132] have shown that NEBBF had 11.7–29.5%, 32.9–64.0% and > 31.4% higher N, P and K utilization rates than conventional fertilizers, respectively. Das and Ghosh [145] demonstrated that NEBBF reduced the downward movement of N and K but enhanced the movement of P in the soil column experiments; after 90 days of leaching, NO_3^- was released by 55.47–50.84%, NH_4^+ was released by 55.47–50.84%, PO_4^{3-} was released by 65.31–68.52%, and K_2O was released by 74.33–77.27%. The nutrient release dynamics of NEBBF mostly relied upon biochar properties, i.e. volatile materials, structure, functional, elemental composition and soil physicochemical attributes such as cation exchange capacity, water content, pH, bulk density, temperature, organic matter content, porosity and specific surface area which in turn augment nutrient retention efficiency [146]. Tailoring NEBBF characteristics, particularly particle size, porosity and specific surface area, is imperative for enhancing adsorption capacity, thereby amplifying soil fertility and consequent crop yields [147–149]. The rate of nutrient release from NEBBF varied substantially across soils with divergent properties, and its prior treatment processes (Table 4), i.e. for acid soils due to ameliorative properties of biochar exchangeable bases/cations (Ca^{2+} , Mg^{2+} and K^+) are released. Ndoung et al. [128] observed maximal potassium release in clay soils, followed by sandy soils and natural silica after a 60-day incubation period. In addition, acidic soils, higher soil temperatures, and higher soil moisture may result in accelerated rates of nutrient release from NEBBF [150–152]. Moreover, NEBBF application exerted indirect effects on crop growth by altering soil physicochemical properties, including soil pH, C content and CEC [153–155]. Notably, NEBBF efficacy is often more pronounced in acidic soils compared to alkaline soils, underscoring the pivotal role of soil pH in mediating NEBBF efficacy [156]. The plausible explanations for this can be originated from different mechanisms such as ligand exchange/electrostatic reactions along with hydrophobic/ π - π processes which are facilitated by interactions of biochar with soil minerals and organic matter. However, the mineral-biochar combinations may improve soil carbon stability and contribute to nutrient biogeochemical cycling. Furthermore, NEBBF may indirectly modulate

inter-root development by regulating soil enzyme activities, microbial abundance, diversity and functional genes [123, 157].

Soil/Plant Effects in Agricultural Applications of NEBBF

Beyond nutrient release, NEBBF application may modulate soil physicochemical and biological properties, thereby exerting indirect effects on crop growth and development [150, 157, 168, 169]. Consequently, the intricate mechanisms governing NEBBF's agricultural application necessitate a comprehensive assessment of its multifaceted impacts (Table 4). Undeniably, nutrient enrichment through NEBBF is pivotal for agricultural sustainability, with diverse enrichment methodologies enabling substantial augmentation of N, P and K content or trace elements to meet crop growth exigencies [134, 170]. Earlier studies typically used feedstocks with high nutrient content to prepare biochar, and more studies now focus on pre/post pyrolysis nutrient enrichment procedures. While most laboratory investigations utilized chemical fertilizers to enrich NEBBF, some studies advocated for further enrichment with mineral fertilizers and organic matter [160]. The multifaceted influence of NEBBF on crop physiology primarily stems from its dual role in nutrient provisioning and amelioration of soil physicochemical attributes (Table 4; Fig. 4). Encouragingly, NEBBF applications have mostly enhanced crop development and yields, with an average of 2.3 times higher crop yields reported [136]. Of concern is that most of the current research on the agricultural application of NEBBF has only compared it to a simple fertilizer control or a biochar control and has not focused on differences in the effects of combined applications. Furthermore, NEBBF application typically enhanced chlorophyll synthesis and photosynthesis, thereby stimulating overall plant growth and development [154]. Insights from various studies indicate NEBBF's potential in modulating enzyme activities, phytohormonal balance, metabolite synthesis and gene expression to regulate plant growth and development [171–173].

The efficacy of NEBBF application may exhibit significant variability across pot and field conditions, highlighting the need for rigorous field evaluations to ascertain real-world efficacy [174–177]. For instance, Haider et al. [178] found significant yield-enhancing effects of biochar under greenhouse conditions but not under field conditions. Climatic and environmental factors, such as rainfall and irrigation, may exacerbate nutrient leaching losses from NEBBF in field settings [179–181], necessitating a holistic appraisal of NEBBF efficacy across diverse environmental conditions. Additionally, NEBBF exhibits significant potential in the remediation of trace metal-contaminated soils by immobilizing heavy metals and reducing their bioavailability and phytotoxicity due to its exceptional adsorption

Table 4 The effects/mechanisms related to agricultural application of NEBBF

Biochar biomass	Enrichment type	Group of feedstocks	Nutrients	Effects/mechanisms of agricultural application	References
Urban green waste	Post-pyrolysis	Chemical fertilizer and clay minerals	N	• NEBBF increased shoot fresh weight by 14% and root fresh weight by 25% in corn compared to the urea control. • Nitrogen was firmly bound to the functionalized surface of NEBBF, resulting in reduced soil nitrogen leaching and effective NH_4^+-N retention.	Shi et al. [158]
Maize stalk, black gram and pine needle	Post-pyrolysis	Chemical fertilizer	N, P and K	• NEBBF significantly increased crop yield by 8.8% as compared to fertilizer application. • After 90 days of leaching, NO_3^- release 55–50%, NH_4^+ 55–50%, PO_4^{3-} 65–68% and K_2O 74–77%. • NEBBF minimized N and K leaching while promoting P mobility. The large surface area and porous structure enhanced NO_3^- retention and enabled gradual nutrient release, though its ability to retain P independently was limited.	Das and Ghosh [145]
Manure	Post-pyrolysis	Gas	S	• NEBBF significantly increased maize plant biomass by 49%. • NEBBF contains high levels (36%) of S and higher concentrations of calcium, potassium, magnesium, manganese, copper, iron and phosphorus and promotes nutrient uptake by the plant. • H_2S adsorption on NEBBF improves soil S fertility by facilitating its conversion to elemental S and SO_4^{2-} through chemisorption, physiosorption or oxidation.	Zhang et al. [159]
Rice straw, maize stover and gram residue	Post-pyrolysis	Chemical fertilizer and organic waste	N	• NEBBF significantly increased the pH and CEC of the acidified soil. • Calcite and dolomite aggregates and surface functional groups in NEBBF helped to neutralize the pH and increase the CEC of the soil. • NEBBF significantly increased soil dehydrogenase, β-glucosidase and leucine aminopeptidase enzyme activities, thereby improving the biological quality of the soil.	Nain et al. [120]
Sewage sludge	Post-pyrolysis	Chemical fertilizer	K	• Radish tuber biomass was significantly increased by 150%. • The rise in K in leaves promoted chlorophyll synthesis and SPAD values. • Slower K release due to low-solubility precipitate formation, electrostatic attraction, micropore action and hydrophobicity. Humic and fulvic acids in charcoal may also complex K, delaying its release.	Fachini et al. [113]
Manure and bamboo	Post-pyrolysis	Chemical fertilizer, organic fertilizer and minerals	N, P and K	• NEBBF significantly increased the nitrogen utilization efficiency of aboveground. • Application of NEBBF improved specific leaf area, photosynthetic efficiency. • NEBBF has the advantage of not limiting nitrogen effectiveness, nitrogen uptake and plant photosynthesis. • Post-heating process of NEBBF enhanced the release of dissolved organic C, improving N availability, and leading to higher foliar total nitrogen (TN) and C assimilation in NEBBF at low rates.	Nguyen et al. [40]

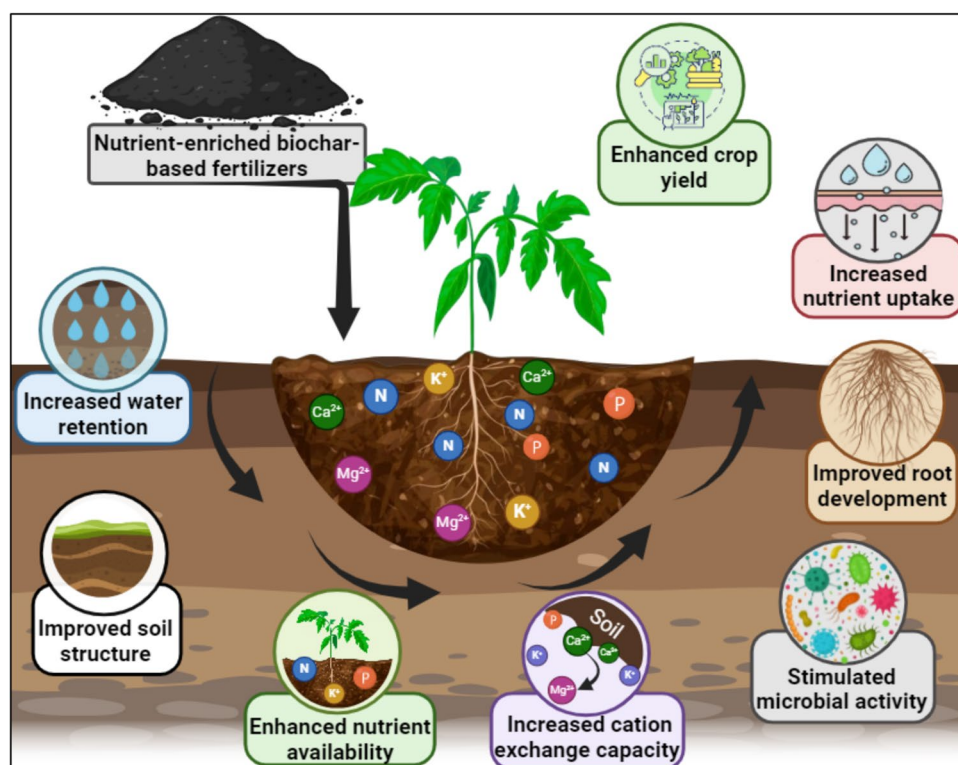
Table 4 (continued)

Biochar biomass	Enrichment type	Group of feedstocks	Nutrients	Effects/mechanisms of agricultural application	References
Waste wood	Post-pyrolysis	Organic waste and compost	N and P	• Maize grain yield increased significantly by 29%. • Maize leaf chlorophyll and above-ground biomass were significantly elevated. • Soil total nitrogen, quick-acting phosphorus, soil water content, soil total organic carbon and cation exchange capacity increased significantly. • NEBBF's accelerates germination and maize growth by modifying soil thermal dynamics, improving water availability and potentially influencing hormones. • The 30% biochar and 70% compost configuration of NEBBF most significantly promoted ryegrass biomass (Biomass growth after three harvests gradually decreased from 150 to 41%). • NEBBF was more effective in stimulating plant growth than N fertilizers. • The beneficial effects of NEBBF on plant were enhanced when combined with N. • NEBBF promotes the mineralization of labile carbon, causing microorganisms to temporarily immobilize nitrogen, thereby raising microbial biomass nitrogen (MBN).	Agegnehu et al. [160]
Waste wood	Post-pyrolysis	Organic waste and compost	N and P	• NEBBF significantly increased cabbage biomass by 375% compared to the blank control. • NEBBF has significantly lower leaching losses of nitrogen and phosphorus than conventional fertilizers. • NEBBF enhances soil carbon and nitrogen metabolism-related bacterial abundance. • NEBBF enhances microbial colonization by providing biochar pores for microbial settlement and adsorption, promoting persistence in soil. This microbial activity regulates soil β-glucosidase, urease and alkaline phosphatase, improving nutrient uptake and boosting biomass.	Lebrun et al. [161]
Straw	Post-pyrolysis	Chemical fertilizer, microorganisms and aubergine soil	N, P, and K	• NEBBF significantly increased cabbage biomass by 375% compared to the blank control. • NEBBF has significantly lower leaching losses of nitrogen and phosphorus than conventional fertilizers. • NEBBF enhances soil carbon and nitrogen metabolism-related bacterial abundance. • NEBBF enhances microbial colonization by providing biochar pores for microbial settlement and adsorption, promoting persistence in soil. This microbial activity regulates soil β-glucosidase, urease and alkaline phosphatase, improving nutrient uptake and boosting biomass.	Wang et al. [162]
Maize straw	Pre-pyrolysis + post-pyrolysis	Chemical fertilizer	N and Mg	• NEBBF significantly increased plant height, aboveground dry weight, root dry weight and chlorophyll content of corn by 20, 24, 29 and 24%, respectively, compared with ammonium nitrate control. • Nitrate and ammonium release rates were about 2.5 and 1.5 times lower with NEBBF compared to the ammonium nitrate control. • NEBBF acts as a hydrogel, enhancing moisture retention with its large surface area and gelatinized polyvinyl alcohol and starch. The slow-release behaviour is shown by higher nitrate release, likely due to cation adsorption on the biochar's negative charges.	Khajavi-Shojaei et al. [163]

Table 4 (continued)

Biochar biomass	Enrichment type	Group of feedstocks	Nutrients	Effects/mechanisms of agricultural application	References
Poultry litter	Pre-pyrolysis	Chemical fertilizer	P and Mg	• NEBBF significantly enhanced maize yield and had a stronger effect in the powder form than in the pellet form. • NEBBF also improves soil pH and magnesium levels, which promotes maize growth. • NEBBF significantly reduced water-soluble phosphorus and acted as a slow-release phosphorus.	Lustosa Filho et al. [164]
Distillers' grains	Pre-pyrolysis	Chemical fertilizer	K	• Fresh weight of <i>Komatsuna</i> was significantly increased by 32% after 25 days. • SPAD values of <i>Komatsuna</i> were significantly increased. • The lowest cumulative release rate of K from NEBBF prepared at CO₂ and 400 °C is regulated by the combination of porous structure and surface functional groups.	Cheng et al. [165]
Rice husk	Pre-pyrolysis	Organic waste	Si	• NEBBF enhanced the salt tolerance of banana by increasing phenolics, carbohydrates and proline in leaf tissue by 35%, 49% and 14%, respectively. • NEBBF significantly reduced MDA content in banana under salt stress. • NEBBF alleviated salt stress by increasing calcium and potassium uptake, improving antioxidant defences and enhancing anti-solvent production.	He et al. [166]
Corn straw	Post-pyrolysis	Chemical fertilizer	N, P and Mg	• NEBBF acted as a slow-release fertilizer, enhancing corn growth and nutrient uptake efficiency. • P-release was regulated by Mg/P precipitate solubility and MgO's "P-trap" effect, while biochar's confinement effect influenced N-release, electrostatic attraction for NH₄⁺ and hydrogen bonding/pore-filling for organic N.	Luo et al. [167]

Fig. 4 Effect of nutrient-enriched biochar on soil health and plant growth



capacity [182–184]. Recent studies have explored NEBBF applications in stressful environments, such as saline and acidic degraded soils, to enhance plant tolerance [120, 166, 173]. Therefore, NEBBF represents a promising avenue for sustainable agriculture, offering controlled nutrient release and enhanced soil fertility (Fig. 4). Nonetheless, elucidating the intricate interplay between NEBBF's slow-release kinetics and crop nutrient requirements necessitates further research. Additionally, comprehensive assessments of nutrient leaching, retention and long-term environmental effects of post-NEBBF application are imperative. Lastly, optimizing NEBBF preparation methodologies to enhance cost-effectiveness and efficiency is paramount to expanding its agricultural applicability.

Environmental Application of Nutrient-Enriched Biochar

Toxic Metal Sequestration by Nutrient-Enriched Biochar Material

The enhanced capacity of biochar to bind and/or sequester toxic heavy metals in contaminated soil environments can be facilitated by modifying it with various materials. The application of P-loaded biochar has been found to significantly reduce the quantity of heavy metals in the soil in addition to raising the P content [185]. Ndoung et al. [134] have reported that the reducible and exchangeable

fractions of heavy metals have been converted into a more stable fraction after the application of nutrient-enriched biochar. Chen et al. [186] have observed that the fertilizers made up of urea-hydrogen peroxide (UHP) and rice husk biochar facilitated the slow release of nitrogen and also contributed to the immobilization of cadmium (Cd) in soil. Islam et al. [187] reported similar results, where it was found that calcium and iron-enriched biochar considerably decreased the level of arsenic (As) and cadmium (Cd) in polluted soils. Gao et al. [188] have observed that coprolyzing biomass with phosphate fertilizer enhanced the carbon sequestration tendency of biochar and promoted the material's capacity for binding toxic metals. They found that the binding capacity of such nutrient-enriched biochar for Cd, Cu and Pb was approximately > 2–4 times higher than that of raw (non-enriched) biochar. Yang et al. [189] have discussed that several processes, such as strong ionic exchange/metal complexation, adsorption (physical/chemical), surface precipitation and electrostatic interaction, govern the heavy metal binding process with nutrient-enriched biochar. Nutrient-enriched biochar exhibits significant benefits not only for the remediation of hazardous compounds and environmental contaminants but also for the harmless treatment of human excreta and the thermochemical conversion of medical waste (Table 5; Fig. 5). Table 5 also justifies the slow-release ability of nutrient-enriched biochar through various enrichment techniques and their subsequent environmental applications.

Table 5 Environmental benefits of nutrient-enriched biochar applications

Feedstock types	Pyrolysis temperature (°C)	Enrichment materials (**)	Environmental applications	Main findings	References
Date palm leaf waste	600	Phosphorus	Mining soil rehabilitation	Phosphorus fertilizer boosted crop growth, increased biomass and aided in stabilizing contaminated areas prone to erosion. It also helped immobilize heavy metals like Pb, Fe, Cu, Zn and Mn in soil.	Ahmad et al. [185]
<i>E. coli</i> fermentation waste	600	Phosphorus	Biomass waste recycling	Converting bacterial biomass into phosphorus-enriched biochar could be a feasible environmental application by recycling fermentation waste as fertilizer.	Kim et al. [190]
Crop residues (rice husks, maize straw or tree pruning)	650–720	Cow urine + dissolved mineral (NPK) fertilizer	Recycling nutrients and biomass carbon	On-farm production of organic biochar-based fertilizers recycles nutrients and biomass carbon, lowering agriculture's environmental impact while increasing small-scale farmer independence.	Schmidt et al. [191]
Forest weed (<i>Eupatorium adenophorum</i>)	450–700	Urine-enriched biochar [manure + mineral fertilizers (P ₂ O ₅ K ₂ O ha ⁻¹ + DAP + MOP)]	Mitigating climate change, food security and poverty	Contribute to addressing the country's present challenges: climate change, food insecurity and poverty.	Pandit et al. [192]
Wheat straw	350–550	Urea with N fertilizer	Reducing greenhouse gas emissions	Applying biochar to dry crop-lands lacking soil organic carbon improved crop yield and reduced greenhouse gas emissions.	Zhang et al. [193]
Coppiced woodlands (birch, beech, oak and hazel)	500	Nitrogen–phosphate and phosphorous fertilizer	Reducing greenhouse gas emissions	Adding biochar to acidic soils enhanced pH without considerably increasing greenhouse gas emissions (CO ₂ and N ₂ O) or decreasing the CH ₄ sink, thus highlighting its potential as a sustainable climate mitigation strategy.	Castaldi et al. [194]
Sugar and distillery industry waste	NA *	NPK fertilizer	Enhancing crop yield and waste management	This enriched biochar acts as a slow-release nutrient source, offering a practical solution for managing waste from the sugar industry.	Eazhilkrishna et al. [195]

Table 5 (continued)

Feedstock types	Pyrolysis temperature (°C)	Enrichment materials (**)	Environmental applications	Main findings	References
Pruning of fig trees and corn cobs	600	NPK fertilizer	Nutrient recycling and sustainable agriculture	Biochar facilitates the slow release of essential nutrients into the soil, diminishing the reliance on synthetic fertilizers and fostering the adoption of organic agricultural practices.	Kizito et al. [196]
Eucalyptus wood	400	Urea and additives (flour, 5% pregelatinized maize and 22% bentonite)	Enhancing crop yield and reducing greenhouse gas emissions	N fertilizers increased the nitrogen utilization efficiency of corn yields, contributing to improved soil carbon sequestration and subsequently reducing net greenhouse gas emissions.	Puga et al. [129]
Aspen, maple, choke cherry and alder (hardwood shavings)	300	Dairy manure effluent	Soil quality improvement and reducing greenhouse gas emissions	Enhancing biochar with dairy manure (effluent) has the ability to enhance soil carbon and nitrogen retention, notably mitigating CO ₂ emissions and decreasing greenhouse gas release.	Sarkhot et al. [197]
Conocarpus wood wastes	400	Poultry manure	Soil quality improvement and carbon sequestration	The utilization of biochar in calcareous sandy soils has the potential to sequester carbon and enhance soil quality by improving nutrient availability.	El-Naggar et al. [198]
Jarrah wood	700	Potassium nitrate, calcium, ammonium phosphate and magnesium sulphate	Mining soil rehabilitation	The combination of jarrah biochar with <i>Acacia</i> helped in restoring mining sites.	Reverchon et al. [199]
Olive mill waste	500	Enriched with nutrients (P and N) and silica	Remediation of heavy metal-contaminated soil	Increased lead (Pb) immobilization in contaminated soil. Augmented nutrient availability. Facilitation of heavy metal-contaminated soil and promotion of soil fertility restoration.	Usama et al. [200]
Brazil nut husk	-	NPK fertilizer	Reforestation/afforestation	Biochar enhanced reforestation/afforestation by boosting plant growth and cutting fertilizer and labour costs.	Lefebvre et al. [201]
Maize silage and biogas residues	500	Potassium (K)	Enhancing nutrient availability (K)	Enhanced the availability of K in a slow-released form.	Wu et al. [202]
Straw and wheat shell	500–600	MgNH ₄ PO ₄ ·6H ₂ O (struvite)	Sustained nutrient release	Extended duration of nutrient release compared to conventional chemical fertilizers.	Hu et al. [203]

Table 5 (continued)

Feedstock types	Pyrolysis temperature (°C)	Enrichment materials (**)	Environmental applications	Main findings	References
Maize stover, rice straw and gram residues	400	Compost	Improving soil quality	Enhanced the biological quality and promoted carbon and nitrogen cycling in acidic tropical soil.	Nain et al. [120]
Softwood chips (spruce-pine-fir) and poultry litter	400, 500 and 600	Ammonium nitrate and urea ammonium nitrate	Reducing nitrogen pollution and promoting sustainable agriculture	Reducing nitrogen leaching and gaseous losses from dairy and poultry manure by effective nitrogen removal.	Jassal et al. [204]

*NA not available

**NPK nitrogen (N), phosphorus (P), and potassium (K); MgNH₄PO₄·6H₂O magnesium ammonium phosphate hexahydrate (struvite), DAP diammonium phosphate and MOP muriate of pot-ash

The unique characteristics of biochar, i.e. porous structure, abundance of mineral components along with functional groups and specific surface area, also aid in the removal of heavy metals (Table 5). From the Table, it is clear that the biochar/struvite composite applications as a promising slow-release fertilizer in soil not only improves nutrient availability (N, P and Mg) and soil conditions for agricultural operations but also offers a “win-win” strategy for producing mineral–biochar composites by eliminating N, P and Mg contaminants from wastewater. The gradual release of nutrients from the biochar/struvite composite is further enhanced by the presence of N, P and Mg as stable struvite minerals in biochar. Researchers have discussed the benefits of applying pyrolyzed carbon for farming activities compared to their non-pyrolyzed component elsewhere [185–188]. The three stages, such as the non-specific and/or selective adsorption stage and the transformation stage, usually drive the immobilization of toxic metals with NEBBF [205, 206]. In general, the heavy metal ion gradually diffuses and penetrates the NEBBF’s micropores during the transformation stage until the equilibrium between the solid and liquid phases is reached. Chemical adsorption is usually considered crucial in the final stage, known as specific adsorption, which leads to the formation of stable complexes, where the heavy metal ions coordinate with the functional groups (organic/inorganic) and either precipitate or co-precipitate with different metallic oxides (Fe/Mn) [206, 207].

Mechanisms Governing the Sequestration of Nutrients

The significant adsorption capacity of biochar for nitrate, ammonium and phosphate and its nano and mesoporous structure make it a good transporter of nutrients like N, P and K [208]. Phosphate adsorption by several types of NEBBF has been discussed by several researchers [48, 196, 209]. The adsorption of nutrients onto biochar/montmorillonite composites revealed that PO₄^{3−} adsorption was regulated by the ionic bonding between PO₄^{3−} and surface functional groups of biochar composites, whereas NH₄⁺ adsorption was mostly attributed to the surface adsorption linked with the ion exchange process. It has been shown that precipitation, ion/ligand exchange, surface metallic complexation and electrostatic interaction caused N to adsorb onto biochar [48, 196, 206]. Wang et al. [210] have shown that surface precipitation, ligand exchange and electrostatic adsorption were the main mechanisms for phosphate adsorption by coal gangue-derived biochar composites. Zhang et al. [210] used goethite/corn straw biochar composites to recover phosphate from synthetic urine, and they found that phosphate adsorption occurred through the formation of inner-sphere complexes (Fe-O-P) and α-FeOOH linkages. Luo et al. [167] have observed that Mg-enriched biochar fertilizer effectively recovers N and P nutrients from the biogas effluent.

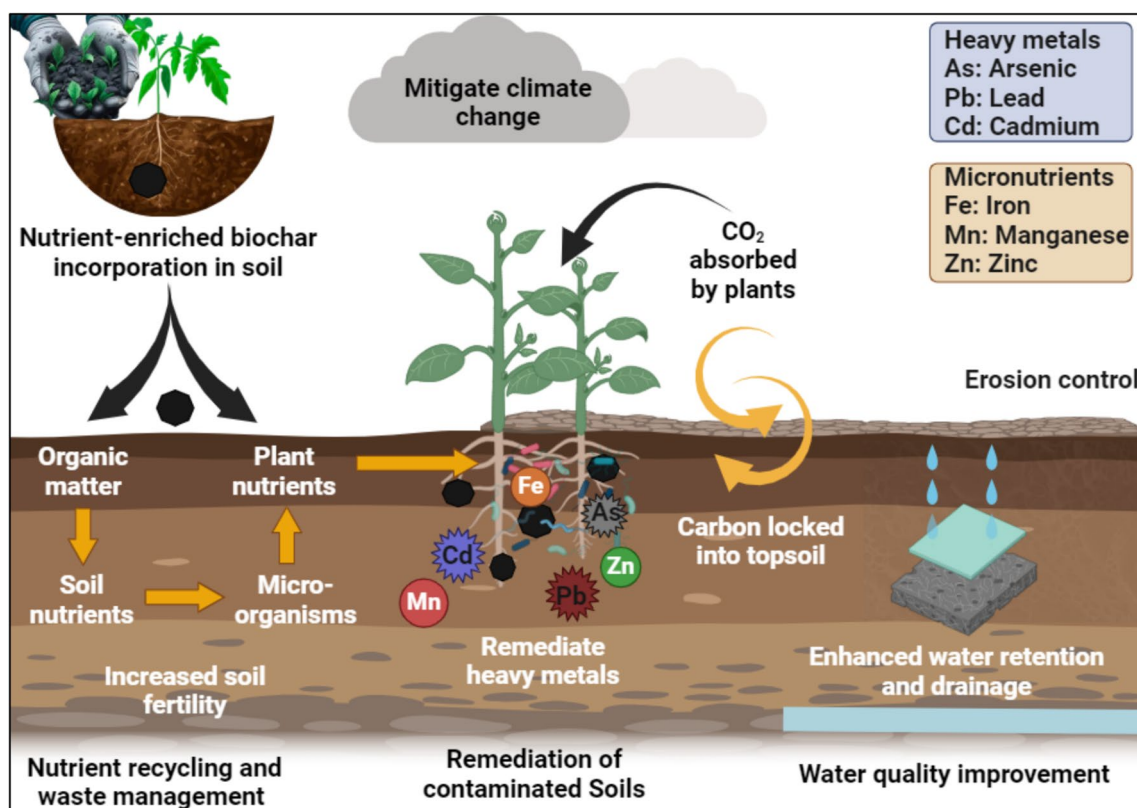


Fig. 5 Effect of nutrient-enriched biochar on environment

P release is connected to the Mg-P precipitates that develop on the surface of the biochar and MgO's "P-trap" action via the PO_4^{3-} precipitation process. For N-containing organic materials, the pore-filling and hydrogen bonding processes dominated the N release; for NH_4^+ , it was dominated by the electrostatic attraction and physical ion trapping effect.

GHG Mitigation by Nutrient-Enriched Biochar Material

Researchers have found that NEBBF is a useful material to curb greenhouse gas (GHG) emissions [134, 211]. Kottegoda et al. [212] have discussed that NEBBF (slow-release fertilizer) made from urea, $\text{NH}_4\text{H}_2\text{PO}_4$, rice straw biochar, and K_2SO_4 was able to considerably reduce greenhouse gas (methane) emissions by > 30% as compared to other conventional chemical fertilizers. Additionally, the researchers have found that during the rice growing stage in the tillering stage, activities of the soil methanogens decreased by > 20%, and from the regreening to filling stages, CH_4 oxidation increased by 14.1%, 52.2% and 27.8%, respectively [213, 214]. These might be attributed to the nitrate (NO_3^-) content of the NEBBF, which might enhance the number and activities of the NO_3^- reducing bacterial population, decreasing the abundance of soil methanogens. Kumar et al. [214] also discussed that NEBBF (nitrogen enriched) mostly

contains stable carbon, which is resistant to microbial degradation and can enhance soil C storage.

Researchers have investigated the correlation between soil nitrous oxides (N_2O) emissions and relevant ecological factors in a subtropical Moso bamboo (*Phyllostachys edulis*) forest ecosystem in China [215]. They found that the application of chemical fertilizers blended with wheat straw biochar reduced the N_2O emissions to the level of > 40%. Due to the large surface area and high CEC of such biochar-blended fertilizer, it may adsorb labile forms of N that might cause a decrease in N_2O emissions, which might also result from the reduced enzymatic activities (governs N-cycling) and water-soluble organic nitrogen concentrations in soil [216, 217]. Shaukat et al. [218] have discussed that lowering soil N_2O effluxes might also be due to the limiting nitrogen supply to the denitrifying bacterial communities.

Effect of Nutrient-Enriched Biochar on Soil Biota and Soil Carbon

The soil microbial biomass, bacterial diversity and abundance could be impacted/promoted by the nutrient-enriched biochar in karst calcareous soils, as discussed by Ndoung et al. [134], indicating that this material had reclamation properties for the karst soils. Furthermore, it was found

that the microbial correlation network developed in size and complexity, as seen by the increased nodes, linkages and rhizosphere nodules in the cladogram produced by the LEfSe analysis. Also, the authors have discussed that the addition of NEBBF led to an expansion of the fungus community in the soil. Chew et al. [119] and [219] observed that the rhizosphere of rice plants when fertilized with activated biochar fertilizer derived from wheat straw pyrolysis, had a high concentration of fungi and bacteria that promote plant growth. Simultaneously, it was observed that twenty-nine fungal functional taxonomic units responded favourably to the addition of NEBBF, whereas twenty-five responded negatively. Additionally, improvement in soil bacterial communities/populations and increased mycorrhizal colonization of wheat roots had been observed after applying a biochar-mineral complex (BMC), i.e. biochar/magnetic iron nanoparticles, compared with the conventional chemical fertilizers [217, 220]. The high porosity of biochar provides changes in the microbial populations because microorganisms occupy pores present in the biochar surface and get protected from the predators, which in turn allow them to proliferate (and grow) in density and variety [213, 214, 221]. Thus, fertilizers made up of biochar can serve as habitats for soil microbes. Researchers have discussed that some nutrients can be transformed by pyrolysis (during biochar production), making NEBBF material suitable for microbial metabolic processes [48, 167, 214].

NEBBF has been found to alter several basic soil processes due to the changes in microbial populations (Table 5). Genome sequencing results showed that nitrifying bacterial communities were dominant while denitrifying bacterial populations were reduced to < 20% after applying NEBBF to the soil [134, 206, 211, 213]. Researchers have demonstrated that a high C input of NEBBF promotes organic matter mineralization and accelerates soil bacterial development, whereas high N input increases the number of nitrifying bacteria [209, 216, 222]. Therefore, this may be the driving factor for the increased nitrifying bacterial populations under the soil treated with nitrogen-enriched biochar. Additionally, saline soil treated with biochar-based fertilizer showed enhanced basal respiration and nitrifying bacterial population [207, 213, 223, 224]. Furthermore, biochar-based fertilizers can stay in the soil for a longer period (decadal time scale) due to their extremely recalcitrant character, which encourages the build-up of soil organic matter (SOM) (Fig. 5). According to Crombie et al. [223], this accumulation is due to the adsorption of the SOM on the surface of the biochar, which serves as a barrier against microbial breakdown of SOM. Several researchers in short incubation experiments (90–120 days) have found that biochar blended with organo-mineral fertilizers provided stabilization of labile carbon in the soil compared to the compost alone and/or biochar-compost mixture [196, 211, 216, 222]. Zhou et al.

[224] found that the application of biochar blended with NPK fertilizer increased the amount of SOM and nutrients in a karst area of south-western China. Zhao et al. [9] reported similar findings where soil organic matter was increased after applying P-enriched biochar in the sandy loam soil.

Conclusions

The concept of utilization of biochar for sustainable agriculture has gained a lot of attention in the last decade. There are several techniques (pre- or post-pyrolysis) to enrich biochar with nutrients to produce biochar-based fertilizers. Chemical adsorption usually drives the coordination of the heavy metals with the biochar functional groups and precipitate/co-precipitate with different metallic oxides due to various chemical interactions. Biochar-based fertilizers not only change the growth of bacterial communities but also change the functions of the microorganisms by upregulating genes associated with the energy production and conversion, transport and metabolism of amino acids, nucleotides, carbohydrates, the translation process and regulation of other genes. Nitrogen-enriched biochar fertilizers (51% biochar blended with 10% nitrogen) tend to lower greenhouse gas emissions (14% less than that of urea treatments).

With the increasing interest in the use of carbon-based fertilizers and their subsequent usages in providing the nutrient requirements of crops and considering the lack of information on the field application and fate of nutrient-enriched biochar fertilizer products, it is important to examine:

- The impact of various feedstock materials and optimization of pyrolyzing conditions to increase the nutrient contents of pristine biochar and assess the CO₂ sequestration potential
- The development of cost-effective synthesis/methods (life cycle analysis based on C emission factors) for the large-scale production of nutrient-enriched biochar products based on the different residual minerals and ground rocks
- The nutrient release characteristics (controlled release behaviour), moisture retention, the bioavailability of mineral nutrients from nutrient-enriched biochar products and the probability of negative priming events
- The quantification of the nutrient use efficiency, crop growth and biomass yield of nutrient-enriched biochar-based fertilizer products under varying agroclimatic conditions, soil types and crop varieties
- The long-term effect of soil application of nutrient-enriched biochar on the fate of C and P's recycling and their role in advancing sustainable agriculture

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Compliance with Ethical Standards

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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